Dominion Energy Virginia
North Anna Power Station
1022 Haley Drive, Mineral, VA 23117
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10CFR50.73

Virginia Electric and Power Company
North Anna Power Station
1022 Haley Drive
Mineral, Virginia 23117

February 26, 2026

Attention: Document Control Desk
U. S. Nuclear Regulatory Commission
Washington, DC 20555-0001

Serial No.: 25-281A
NAPS: RAP
Docket Nos.: 50-338
License Nos.: NPF-4

Dear Sir or Madam:

Pursuant to 10CFR50.73, Virginia Electric and Power Company hereby submits the following Licensee Event Report applicable to North Anna Power Station Unit 1.

Report No. 50-338/2025-002-01

This report has been reviewed by the Facility Safety Review Committee and will be forwarded to the Management Safety Review Committee for its review.

Sincerely,

James Jenkins
Site Vice President
North Anna Power Station

Enclosure

Commitments contained in this letter: None

cc: J. Lara – NRC Region II Administrator Region II
NRC Senior Resident Inspector – North Anna Power Station

NRC FORM 366 (04-02-2024) — U.S. NUCLEAR REGULATORY COMMISSION

LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

APPROVED BY OMB: NO. 3150-0104 — EXPIRES: 04/30/2027

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name		2. Docket Number	3. Page
North Anna Power Station	■ 050 / ☐ 052	00338	1 OF 3

4. Title
Unit 1 Automatic Reactor Trip Due to PRNI High Negative Rate

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	☐ 050	Docket Number
10	21	2025	2025	- 002 -	01	02	26	2026	Facility Name	☐ 052	Docket Number

9. Operating Mode	10. Power Level
1	74

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	**10 CFR Part 50**	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☑ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	**10 CFR Part 21**	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	**10 CFR Part 73**	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ **OTHER** (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include area code)
James Jenkins, Site Vice President	(540) 894-2101

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
X	AA	CON	W893	Y					

14. Supplemental Report Expected		15. Expected Submission Date	Month	Day	Year
☑ No	☐ Yes (If yes, complete 15. Expected Submission Date)				

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On October 21, 2025, at 2108 hours with Unit 1 in Mode 1, 7 4% power, an automatic reactor trip occurred due to power range nuclear instrument high negative rate. The cause of the negative rate was due to control rod C7 dropping into the core. Control rod C7 dropped due to a relaxing of pin prongs that caused the Stationary Coil circuit to open. A 4-hour report was made per 1 O CFR 50.72(b)(2)(iv)(B) for a Reactor Protection System (RPS) Actuation and an 8-hour report was made per 10 CFR 50.72(b)(3)(iv)(A) for a valid Engineered Safety Feature actuation for the Auxiliary Feedwater (AFW) Pumps actuating as designed during the event.

Unit 2 was not impacted by this event. This event is reportable per 10 CFR 50.73(a)(2)(iv)(A) for RPS actuation and valid AFW actuation. The health and safety of the public were not affected by this event.

LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

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1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER		
			YEAR	SEQUENTIAL NUMBER	REV NO.
North Anna Power Station	■ 050 ☐ 052	00338	2025	- 002 -	01

NARRATIVE

1.0 Description of Event

On October 21, 2025, at 2108 hours with Unit 1 in Mode 1, 74% power, an automatic reactor trip occurred due to power range nuclear instrument (EIIS Component DET, System IG) high negative rate. The cause of the negative rate was due to control rod (EIIS Component ROD, System AA) C7 dropping into the core. The Auxiliary Feedwater (AFW) Pumps (EIIS Components P, System BA) actuated as designed during this event. The reactor trip and AFW pumps starting are reportable per 10 CFR 50.73(a)(2)(iv)(A) as System Actuations.

The direct cause of this event was the loss of continuity to the stationary gripper coil for control rod C7. The cause of C7 dropping into the core was a relaxing of the pin prongs resulting in reduced pin spread on the "E" Pin. This resulted in a poor connection on the plug side of the Patch Panel connection inside of containment. The poor connection caused the Stationary Coil circuit to open. This resulted in the Stationary Coil losing its holding current. The Stationary Gripper then opened, dropping C7. C7 dropped from 233 to 183 steps, and this created a change in flux that was picked up by nuclear instruments N41 and N43. This satisfied the 2/4 NI High Negative Flux Rate Trip, which opened the Reactor Trip Breakers (EIIS Component BKR, System AA).

2.0 Significant Safety Consequences and Implications

No significant safety consequences resulted from this event due to plant equipment performing as designed. The health and safety of the public were not affected by this event.

3.0 Cause of the Event

The direct cause of this event was the loss of continuity to the stationary gripper coil for control rod C7. The root cause is that risk reduction corrective actions from the 2024 Unit 1 K4 rod drop did not drive U1 CRDM patch panel replacement and did not prevent reoccurrence of a similar failure since a root cause evaluation was not performed.

4.0 Immediate Corrective Action

The plug for control rod C7 was replaced.

LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

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	☐ 052				

NARRATIVE

5.0 Additional Corrective Actions

The Post Trip Review procedure will be revised to include prior to unit restart; a requirement to identify the extent of condition and to ensure corrective actions are taken to address extent of condition, or an assessment of the operational risk with uncorrected issues shall occur and ensure actions to address the extent of condition are in place. The Work Order Planning procedure will be revised to require use of the procedure Control of Fabricated and Modified Tools, when a craft-made tool is used. The Control of Fabricated and Modified Tools procedure will be revised to require training needs analysis when a craft-made tool is used. A Lessons Learned Case Study will be created for Process and Communication Weaknesses found regarding the craft-made plug/socket gage tools creation/use during NAPS U1 CRDM patch panel inspections without the use of the procedure Control of Fabricated and Modified Tools. Additionally, the electrical maintenance procedure for connecting and disconnecting the Control Rod Drive Mechanism (CRDM) cables is being enhanced.

6.0 Actions to Prevent Recurrence

Complete the work order to splice U1 CRDM patch panel connections during N1R32 (Spring 2027). The existing U1 CRDM patch panel connector will be upgraded to eliminate split plug pin design in N1R33 (Fall 2028). Additionally, the Corrective Action Program procedure has been modified to require a root cause evaluation for all unplanned reactor trips.

7.0 Similar Events

A similar event occurred in May of 2024 at North Anna, LER N1-2024-001.

8.0 Additional Information

Unit 2 was not impacted by this event. The health and safety of the public were not affected by this event.